PHYS 6031
Spring 2024
Homework 1
Due: Friday 26 January 2024

1. Evaluate the *gradient* of the following scalar functions, in the specified coordinate system:

(a) $f = x^2 y^3 z^4$ (cartesian)
(b) $f = \cos\phi/\rho$ (cylindrical)
(c) $f = r\cos\theta$ (spherical)

2. Evaluate the *divergence* and *curl* of the following vector functions, in the specified coordinate system:

(a) $\vec{\boldsymbol{f}} = x^2 y^3 \hat{\boldsymbol{\imath}} + z^4 \hat{\boldsymbol{k}}$ (cartesian)
(b) $\vec{\boldsymbol{f}} = \hat{\boldsymbol{r}}/r^3$ (spherical)

3. Using Dirac delta function in the appropriate coordinate system, express the following charge distributions as three-dimensional densities $\rho(\vec{\boldsymbol{x}})$.

(a) In spherical coordinates, a charge Q uniformly distributed over a spherical shell of radius R.
(b) In cylindrical coordinates, a uniform surface charge distributed on an infinite cylinder of radius b. The charge per unit length along the length of the cylinder is λ.
(c) In cylindrical coordinates, a charge Q spread uniformly over a flat circular disk of negligible thickness and radius R.
(d) In spherical coordinates, a charge Q uniformly spread around a ring of radius R that lies in the $x - y$ plane.

4. An electric potential is given by

$$\Phi = \frac{q}{4\pi\epsilon_o}\frac{e^{-\alpha r}}{r}(1 + \frac{\alpha r}{2})$$

where r is the radius in spherical coordinates, q is the magnitude of the electronic charge, and $\alpha > 0$ is a constant. Find the distribution of charge (both continuous and discrete) that will generate this potential and interpret your result physically. [Hint: You might check on what is the total charge of the system.]

5. Which of the following two expressions can be an electrostatic field? Why or why not? (C is a constant which has the appropriate units.) If it is a valid $\vec{\boldsymbol{E}}$ field, determine the potential function using the origin as the zero reference of the potential.

(a) $\vec{\boldsymbol{E}}(x, y, z) = C\sqrt{x^2 - y^2 + 2z^2}\,\hat{\boldsymbol{z}}$
(b) $\vec{\boldsymbol{E}}(r, \theta, \phi) = Cr(\cos^2\theta\,\hat{\boldsymbol{r}} - \cos\theta\sin\theta\,\hat{\boldsymbol{\theta}})$